Dang Khoa Huynh

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Summary

Third-year mechatronics and space engineering student who designs, builds and tests flight and ground electronics. Primary electronics engineer on an in-situ resource utilisation (ISRU) payload that won the SDL Payload Challenge at the 2026 International Rocket Engineering Competition. Now co-leading a student rocket's ground systems.

Education

University of Sydney – BEng (Honours), Mechatronics with Space Engineering 2024 – 2028 (expected)

- Third year, WAM 90.3, Dalyell Scholar, Dean's List 2025, Robotics and Intelligent Systems specialisation
- Units: Mechatronic Systems Design, Mechatronics 2, Space Engineering 1 & 2, Electronic Devices & Circuits

Skills

Electrical design: schematic/PCB layout (Altium, KiCad), MOSFET drivers, circuit protection, SPICE
Build and test: SMD soldering and PCB assembly, bring-up and test firmware, oscilloscope debugging
Embedded: STM32 (H7, F3), ESP32-S3, FreeRTOS, ARM assembly, SPI/I2C/UART/CAN/RS-485, WiFi/BLE
Programming and CAD: C/C++, Python, MATLAB, ROS2, Git, Linux, Fusion 360, SolidWorks, Onshape
Engineering practice: requirements management, design reviews, technical reporting

Experience

Ground Support Equipment (GSE) Co-Lead, USYD Rocketry Team July 2026 – Present

- Lead ground-systems electronics and software (launch controller, pad box, antenna tracker, umbilical). Defined the pad-box redesign as five team-designed boards on point-to-point RS-485 (Modbus RTU) links
- Designed the 12-channel Valve Control Board (STM32H755, 4-layer, Altium): 7 A PWM MOSFET channels, opto-isolated gate drive, per-channel INA237 current sensing, fused and TVS-protected rails
- Designed the Umbilical & QD Board: USB host to both flight computers, ideal-diode input protection with under/overvoltage lockout, relay-driven quick disconnect and payload charging
- Wrote the design briefs new engineers used for three of the boards, the team's PCB design guide and induction task, and led the electronics section of the Critical Design Review

Payload Primary Electronics Engineer, USYD Rocketry Team – Rock Daisy Payload 2025 – 2026

- **1st place, SDL Payload Challenge, International Rocket Engineering Competition 2026** (150 teams) for a CubeSat-class ISRU experiment on electrostatic regolith separation (USYD news)
- Designed the PCB (three revisions) and firmware for the dual-ESP32-S3 flight computer and OV5640 camera node: 20 MHz SPI sensors, SD logging, actuators and interlocked 7 kV switching
- Architected a five-task FreeRTOS system with a launch-triggered flight state machine
- Built a WiFi checkout dashboard for bring-up and pre-flight testing: actuate each component individually, read live telemetry and board health, and pull logs from the SD card
- Assembled the flight PCB and designed and built the payload's internal cable harness

Low Voltage Engineer, Sydney Motorsport (FSAE Electric) 2025 – 2026

- Redesigned the tractive-system status light on a 12 V aluminium PCB with 700 mA power LEDs, and verified daylight visibility against competition rules with a photometric analysis
- Designed a test motherboard for integration testing of the brake-system plausibility (BSPD) boards, which monitor pedal travel, and the control circuits that switch the vehicle's 480 V accumulator

Projects

2-Axis Linear Motion Stage with Lidar Obstacle Avoidance Bare-metal C, STM32F3

- Designed the electronics, cable harness and 3D-printed rail, base and carriage parts for a two-axis stage
- Wrote interrupt-driven bare-metal C (no HAL) driving two stepper motors via TMC2209 drivers, with serial keyboard control and lidar-based wall avoidance (RPLIDAR A1M8)